Camping Tigers, Hiking Dragons: Shallow Processing of Dangling Modifiers

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In English, when the subject of the matrix clause is incongruent with its present participial verb modifier, the result is a construction known as a "dangling modifier". For example, the sentence *Driving on the way to the store, a deer darted across the road,* may be considered unacceptable because the subject, *a deer*, is incapable of performing the action described by the participial verb *driving*, thus leaving the modifier "dangling". However, this prescriptivist decree of ungrammaticality operates on the assumption that the language processor tries immediately to integrate *deer* with *driving* and checks their compatibility. Previous literature in formal linguistics has also suggested that the processor initiates a search for a suitable controller upon encountering such modifiers, starting with the matrix subject [1, 2].

The current study examined these claims with an eyetracking-while-reading experiment and a stop-making-sense task. Thirty-six experimental item sets were constructed as in Table 1. The design crosses the factors Modifier Type (PP vs. VP) and Subject Type (Human vs. Animal). Modifier Type refers to whether the subordinate clause before the matrix clause is a PP (a, b) or VP (c, d). In the VP condition, the verb always describes an action that can only be performed by humans (*driving*, *reading*, etc.) and is the head of the same PP in the PP condition. The factor Subject Type refers to whether the subject of the matrix clause is human (b, d) or animal (a, c). Within items, the word length of the matrix subject and matrix verb were the same.

Exp 1. In the eyetracking experiment, 40 participants read the experimental sentences mixed with 60 filler sentences one at a time. Log-transformed reading times (first pass duration and regression path duration) were analyzed at the subject region and the verb (spillover) region. If the processor attempts to integrate the subject with the modifier, then the VP + Animal condition would either fail the compatibility check or force an implausible interpretation, presenting larger processing difficulty and resulting in longer reading times. However, such an effect was not observed in the data. Linear mixed effects models failed to find a significant interaction between Modifier Type and Subject Type for both first pass duration and regression path duration at both subject and verb regions (Figure 1).

Exp 2. In the stop-making-sense task, 120 participants read the sentences one word at a time and pressed a button when they judged that the sentence no longer made sense. The percentage of trials where the participant pressed the button were analyzed. We observed a significant interaction between Modifier Type and Subject Type at both subject (β=-1.18, p = .02) and verb regions (β = -2.00, p < .001). Participants were more likely to indicate that the sentence stopped making sense at the matrix subject or matrix verb when the sentence type was VP + Animal (Figure 2). Notably, despite the semi-offline nature of the task substantially increasing anomaly detection, more than 75% of dangling modifiers slipped under the radar at the matrix verb, and around 40% remained undetected even after participants had read the entire sentence.

As a post-hoc analysis, the cosine similarity between the present participial verb in the modifier (e.g., *driving*) and the matrix subject (e.g., *deer*) was calculated using BERT contextual embeddings to see if semantic compatibility could explain item-level variation in the detection rate of dangling modifiers in Exp 2. A binomial mixed-effects model revealed a significant main effect of cosine similarity at the matrix verb region (β = -2.02, p = .008): higher semantic compatibility was associated with a lower probability that participants judged the sentence as no longer making sense (Figure 3). No corresponding effect was found at the subject region (β = 0.94, p = .35).

The results suggest that dangling modifiers do not incur immediate processing costs during natural reading, consistent with a surprisal-based processor that does not register the anomaly unless semantic irregularity is sufficiently strong. Even when readers are forced to closely monitor sentence coherence, a substantial proportion of dangling modifiers remain undetected. However, when detection does occur, it happens quickly and detection rate is systematically modulated by semantic compatibility. In sum, the interpretation of dangling modifiers is a delayed and semantically driven evaluation, rather than an obligatorily early attempt to establish syntactic control relations during incremental language processing.

(a) On the way to the store, a deer darted across the moonlit road. (PP, Animal)
(b) On the way to the store, a teen turned left in the wrong lane. (PP, Human)
(c) Driving on the way to the store, a deer darted across the moonlit road. (VP, Animal)
(d) Driving on the way to the store, a teen turned left in the wrong lane. (VP, Human)
Table 1. Example set of experimental sentences.

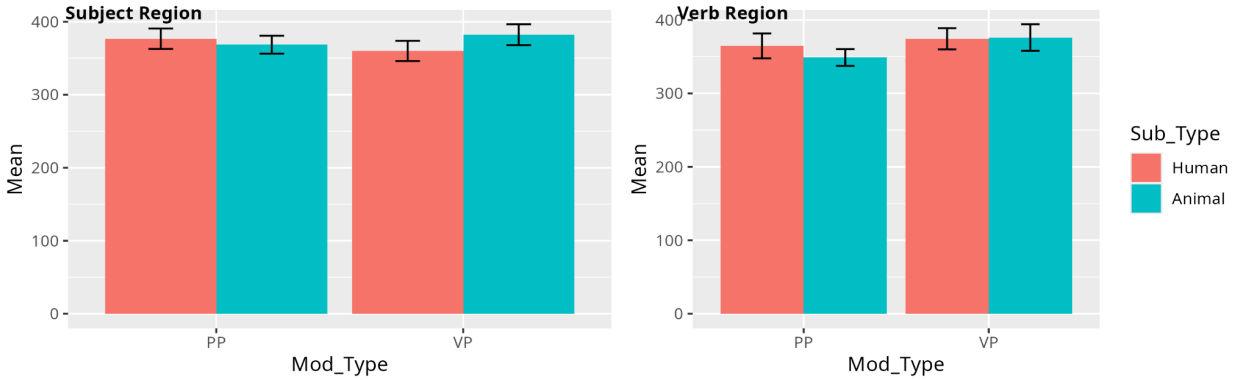

Figure 1. Average regression path durations in subject and verb regions in Exp 1. Error bars represent 95% CI.

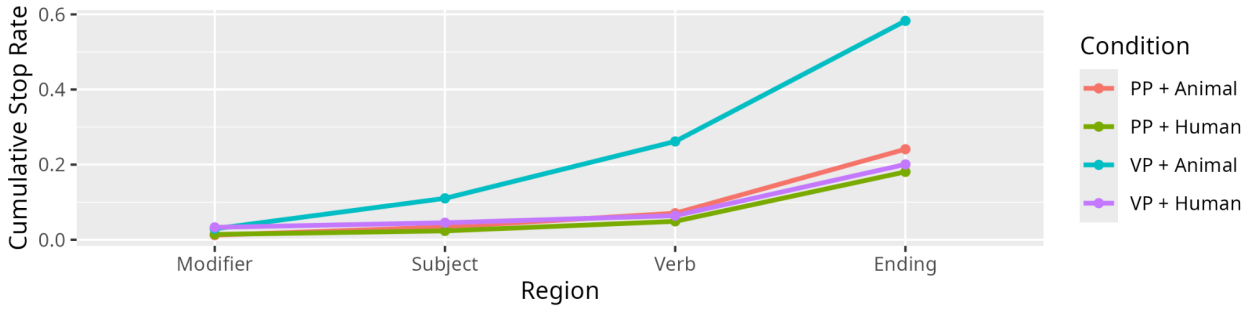

Figure 2. Cumulative percentage of trials in Exp 2 where participants indicated that the sentence stopped making sense in each condition by sentence region.

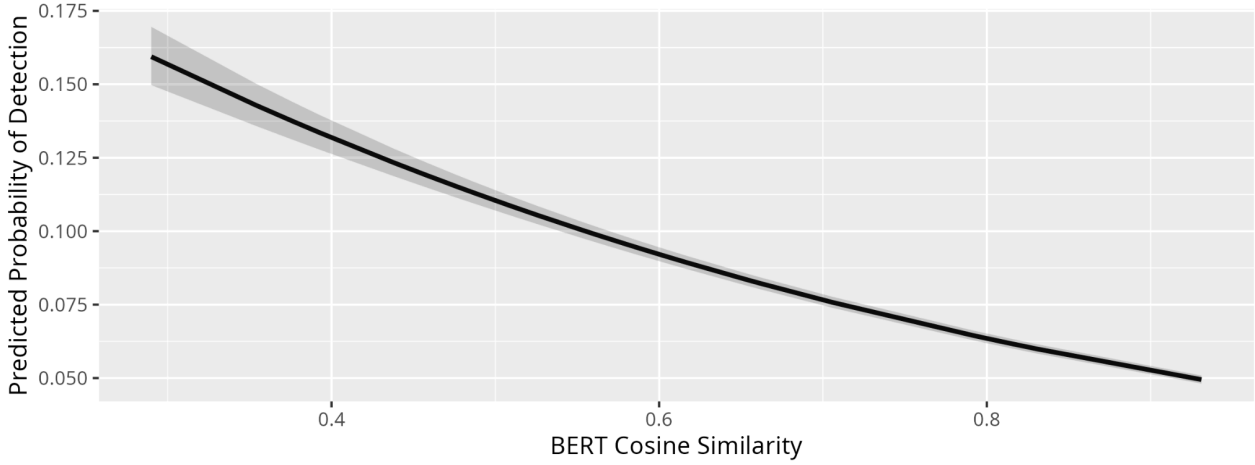

Figure 3. Predicted probability that a dangling modifier “stops making sense” at the matrix verb region as a function of the semantic similarity between the participial verb and matrix subject.

[1] Donaldson, J. (2021). Control in free adjuncts: the dangling modifier in English.
[2] Kortmann, B. (1991). Free Adjuncts and Absolutes in English: Problems of Control and Interpretation